

## 1 The Take-away

- Temporal adverbs do not directly presuppose about *p*
- But rather relate the topic time to a change situation
- Evidence from Kiowa (kio)

## 2 Kiowa /hègɔ́/ wears many time hats

The adverb /hègɔ́/ is the most commonly used word in the language, as it applies to a wide array of temporal meanings, leading to a number of various translations in English.

[ INV = inverse number, which with inanimates is singular (Watkins 1984, Harbour 2008). A: agent, O: object, S: unaccusative subject, D: dative/applicative ]

|               |                                         |              |           |                          |              |              |
|---------------|-----------------------------------------|--------------|-----------|--------------------------|--------------|--------------|
| (1) ‘already’ | /á:dò                                   | dè=          | tʰón/     | /hègɔ́                   | è=           | tʰêm+dò:/    |
|               | stick:INV                               | lSGA:3INV O= | find.PFV  | <b>already</b>           | 3INV S=      | break+be     |
|               | ‘I found a stick.’                      |              |           | ‘It was already broken.’ |              |              |
| (2) ‘then’    | /á:dò                                   | dè=          | tʰón/     | /hègɔ́                   | dè=          | tʰêm/        |
|               | stick:INV                               | lSGA:3INV O= | find.PFV  | <b>then</b>              | lSGA:3INV O= | break.PFV    |
|               | ‘I found a stick.’                      |              |           | ‘Then I broke it.’       |              |              |
| (3) ‘now’     | /á:dò                                   | dè=          | tʰêm/     | /hègɔ́                   | è=           | têm+dò:/     |
|               | stick:INV                               | lSGA:3INV O= | break.PFV | <b>already</b>           | 3INV S=      | break+be     |
|               | ‘I broke a stick.’                      |              |           | ‘Now it is broken.’      |              |              |
| (4) ‘no more’ | /á:dò                                   | nó=          | dó:/      | /hègɔ́                   | hón          | nó=          |
|               | stick:INV                               | lSGD:3INV S= | be        | <b>no more</b>           | not          | lSGD:3INV S= |
|               | ‘The stick was mine.’                   |              |           | ‘It isn’t anymore.’      |              |              |
|               | This is really just ‘now it isn’t mine’ |              |           |                          |              |              |

In addition to these meanings, multiple distinct uses of /hègɔ́/ can co-occur in a sentence. In (5), the first use of /hègɔ́/ is not translated at all (by an L1 Kiowa speaker who learned English in boarding schools), but describes the temporal setting and we gloss it as ‘now’. The second use, translated as ‘already’, applies to the time having passed midnight. The third, translated as ‘so’, is glossed as ‘then’ for consistency, but its translation clearly signals the sequence of spotting the time before preparing to stop. Does /hègɔ́/ encode the causation, or is that simply inferred?

- (5) pàj+èl già= bó: nò hégó gí:+kʰòp'ǝj+thòp hégó Ø=tʰó:gíáj nò  
 clock+big lSGA:3PLO= look at.PFV and now night+mid+past already 3PLS=pass thru.PFV and  
 hégó gíat=pʰátkjá+mò:hòl  
 then lSGA:3PLO=stop+prepare.PFV  
 'I looked at the big clock and it was already past midnight so (then) I prepared to stop.' ('A Ghost  
 Woman Demands a Check', P. McKenzie n.d.)

### 3 Each instance of /hégó/ involves a preceding change

Obviously we have the question: are there multiple /hégó/'s in Kiowa, or is this lexical item's meaning broad enough to cover all of these? A priori the latter, which suggests that the English adverbials share a common core of meaning.

What brings these meanings together? In each case, some relevant change situation precedes the situation described by the /hégó/-clause. For instance, in (1), the change situation was the stick breaking, which happened before the topic time/situation of the clause 'it was broken,' which contains a result-state stative (McKenzie under review)– at the time I found it, it was broken. Since the change preceded the finding, /hégó/ is warranted to express that.

Some of these change situations are more 'definite' than others, warranting pronouns. Whether the less definite cases involve quantification or choice functions remains to be determined.

| clause 1                      | potential change situation | clause 2                    |
|-------------------------------|----------------------------|-----------------------------|
| (1') $s_1$ I found a stick    | $\exists s$ it broke       | $s_1$ it was already broken |
| (2') $s_1$ I found a stick    | $s_1$ I found it           | $s_2$ then I broke it       |
| (3') $s_1$ I broke a stick    | $s_1$ I broke it           | $s_2$ now it is broken      |
| (4') $s_1$ The stick was mine | $s_2$ I lost it            | $s_3$ it isn't mine anymore |

This change situation can be the start of a perfect-time interval, which allows the expression of perfect meanings in the absence of any perfect inflection in Kiowa, whether that involves an experiential or universal perfect.

|                                                                                                           |                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| (6) /jír:-dò France-kù hégó à=bá:/<br>two-times France-to have lSGS=went<br>'I had gone to France twice.' | (7) /hégó háòtè+kʰi: già=kʰînmò/<br>have several+day lSGA:3PLO=be coughing<br>'I've been coughing for several days.' (Watkins 1984: 219) |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|

### 4 Proposal

The core of meaning for /hégó/ is that a change situation (c) precedes a time (i), which is generally the topic time. We can propose this formally:

- (8) a.  $\llbracket \text{hégó} \rrbracket^g = \lambda c \lambda i. g(c)$ , only if  $i \succ \text{time}(g(c))$  type  $\langle s, \langle i, t \rangle \rangle$   
 b.  $\llbracket \text{hégó} \rrbracket^g(s_0) = \lambda i. g(0)$  [ $i \succ \text{time}(g(0))$ ] type  $\langle i, t \rangle$

Applied to a change situation (like  $s_0$ ) in (8b), the result is a property of times for the topic time to saturate.

This meaning is unaffected by negation (as in (4)), or by conditionals, so we can classify it as a presupposition.

## 5 Filling in relations

If /hègɔ́/ describes change situations that precede the topic time, we can look for adverbs that describe other relations. These are not hard to find.

The adverb /hétɔ́/ ‘still’ places the topic time **before** a change situation. It is translated by English ‘still’ or with negation, ‘(not) yet’. In (9), the topic time  $t_1$  is the speech time, but the change situation is in the future.

- (9) é:hò:=àl  $t_1$  **hétɔ́** hón bét= t’á:gjà+òm̃gɔ́:-hèl  
 now=even  $t_1$  **still** not 3INVD:3INVS= right+be made.NEG-HSY  
 ‘even now they have not gotten things straightened out yet.’ [ INV = inverse number, which with animates is triplural ]

Whether this change situation needs to be formalized with modality, as it has not yet occurred, is a matter for later research.

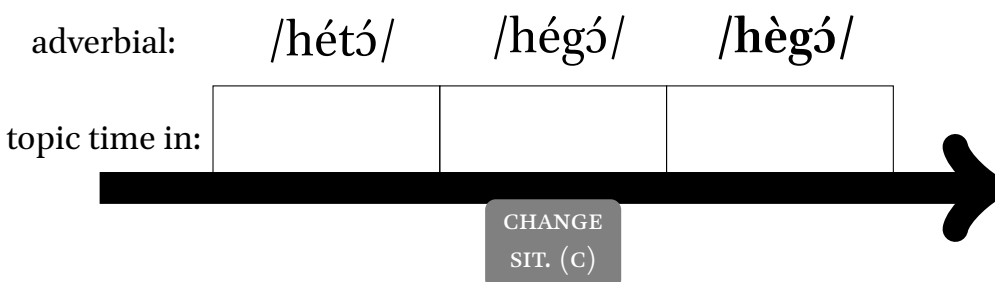
The adverb /hégɔ́/ ‘just as’ places the topic time **during** a change situation. Note the tone difference from /hègɔ́/. In (10), the principal was telling the wayward student while not punishing him for running away from school.

- (10) hègɔ́  $t_1$  hón hájá é= ó:m-ô:.  
 then not at all 3SGA:1SGO= do-NEG  
 ‘Then he didn’t do anything to me.’ (*i.e.* punish)  
**hégɔ́**  $t_2$  é= tó:tó: gò pòj k’í:gjà tsô: bàt= ôj+òm-tò:  
**just as** 3SGA:1SGO= speak to.PFV and not later thusly 2SGA:3PLP= again+do-MODAL  
 ‘And he was telling me, “Don’t do it again.”’

In (11), the narrator is recounting how wide-eyed spectators were mouths agape while they beheld the spectacle of Kiowas butchering a cow.

- (11) tá:gò mîn bét= k’ú:lyà gò **hégɔ́** ét= sòl+k’jàl+dò:  
 eye:INV about to 3INVD:3INVS= pop out:IPFV and **just as** 3INVA:REFLO= mouth+agape+hold  
 ‘Their eyes were about to pop out and they had their mouths open wide.’ (McKenzie et al. 2022: S167)

The figure below demonstrates how these adverbs cover different temporal spaces relative to the change situation. If the topic situation comes before it, it lands in the /hétó/ zone, so that adverb is used. After it, and it lands in the /hégó/ zone, so that adverb is used. Otherwise, /hègó/ is used.



## 6 These resemble non-narrative *now*

Recent proposals argue that non-narrative uses of *now* and *then* in English are not deictic with respect to the utterance time but rather to an assessment time (Anand & Toosarvandani 2019, Stojnić & Altshuler 2021). In (??, the adverbs are either during the assessment time (now) or not (then), even if that time is not the utterance time.

- (12) a. *She was **now** a senator.* (  $d \subseteq \text{time}(a)$  )  
 b. *She **then** becomes a senator.* (  $d \not\subseteq \text{time}(a)$  )

For these accounts, the slip from deictic to narrative is straightforward— they are both deictic, but to different times.

The Kiowa case is different: Deixis to the utterance time requires a complete different set of adverbials, which are also used for spatial deixis.

(13)

| stem | deixis           | with definite | gloss                |
|------|------------------|---------------|----------------------|
| é:   | proximal         | é:hò:         | 'now, at this time'  |
| ó:   | medial/discourse | ó:hò:         | 'then, at that time' |
|      |                  | ójhò:         | 'around then'        |
| ó:   | distal           | ó:            | 'long ago'           |
| ôj   | exact            | ôjhjò:        | 'right now/then'     |

Most of the time, when these stems are used for temporal deixis, they also carry the suffix –hò: 'DEFINITE', as exemplified in (9). This suggests that the change-situation adverbs do not need an assessment time.

## 7 Extension to English *already*

How does /hègó/ compare to its other common English translation, 'already'? Analyses of *already* and similar German adverbs involve presupposing *p* to hold or not hold during an interval related to the topic

time  $t$ . Löbner (1989) argues that *already* presupposes that  $p$  is false before  $t$ . Krifka (2000) and Ippolito (2007) each provide counterexamples like (15) and argue that *already* presupposes that  $p$  is true after  $t$ .

(14) *I am **already** at home.*

(15) (true before  $t$ )

A: *Is your husband applying for citizenship?*

B: *He is **already** American, since he was born there.* (Ippolito 2007)

That said, it is easy to find cases where that presupposition fails.

(16) (false after  $t$ )

*When I wrested the gun from the assailant's hands, he was **already** aiming it at me.*

*She was **already** a senator when the scandal forced her resignation from public life.*

A change-situation analysis fits well, though: In each case, Some relevant change situation precedes the topic situation described by the *already* clause (minus *already*).

|        |             | change situation (in context) | <i>already</i> -clause       |
|--------|-------------|-------------------------------|------------------------------|
| (14')  | $\exists s$ | I got home                    | $s_1$ I am home              |
| (15')  | $\exists s$ | he gained US citizenship      | $s_1$ He is American         |
| (16a') | $\exists s$ | he took aim at me             | $s_1$ he was aiming it at me |
| (16b') | $\exists s$ | she became a senator          | $s_1$ she was a senator      |

## 8 Some avenues to explore

A full account would try to understand what makes us choose particular change situations, or what kinds of implicatures we can generate from the choice.

Also, there may be cases where *already* works with an assessment time rather the topic time/situation.

Finally, while Kiowa uses /hègó/ for many things, English splits these meanings onto distinct adverbials. The change-situation analysis suggests that the core of these is simply that, a change related to the topic time. Thus, any differences between *now*, *then*, & *already* may involve additional presuppositions or conditions.

## References

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